SANT'ANNA WORKSHOP ON CAUSAL INFERENCE

Covariates in DiD

OLS specs, IPW, OR, DR

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Recall: the 2×2 is DiD — six equivalent forms

1. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t] - \mathbb{E}[Y|D{=}1,t{-}1]\big]}_{\text{first difference}} - \underbrace{\big[\mathbb{E}[Y|D{=}0,t] - \mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{first difference}}$
2. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t] - \mathbb{E}[Y|D{=}0,t]\big]}_{\text{group difference}} - \underbrace{\big[\mathbb{E}[Y|D{=}1,t{-}1] - \mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{group difference}}$
3. $Y = \alpha_1 + \alpha_2\,\mathrm{Post} + \alpha_3\,D + \delta\,(\mathrm{Post} \times D) + \varepsilon$ *saturated regression*
4. $Y = \alpha_1 + \tau_t + \sigma_i + \delta\,(\mathrm{Post} \times D) + \varepsilon$ *two-way fixed effects*
5. $\Delta Y_i = \alpha_1 + \delta\,D_i + \varepsilon_i$ *first-difference / horizontal regression*
6. $\Delta \bar{Y}_G = \alpha_1 + \delta\,\mathrm{Post}_G + \varepsilon_G$ *group / vertical regression*

All six are **numerically identical**.

The 2×2 DiD estimand

$$\underbrace{\widehat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\Big[\mathbb{E}[\Delta Y(0) \mid D{=}1] - \mathbb{E}[\Delta Y(0) \mid D{=}0]\Big]}_{\text{PT bias}}$$

Treatment $=$ college education. Outcome $=$ wages.
Parallel trends requires the PT bias term to equal zero.

A thought experiment

Group	Untreated wage trend $\Delta Y(0)$
Male workers	$+10$ per year
Female workers	$+8$ per year

Case 1: $D{=}1$ is 75% male. $D{=}0$ is 75% male.

Will parallel trends hold?

Case 1: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.75 \times 10 + 0.25 \times 8$	9.5

Parallel trends holds? **Yes.**

$\widehat{\delta}_{\text{DiD}} = \text{ATT} \checkmark$

Case 2: unequal composition

	% male
$D=1$ (college educated)	75%
$D=0$ (no college)	25%

Same wage trends as before ($+10$ male, $+8$ female).
Will parallel trends hold now?

Case 2: show the work

	Calculation	**Result**
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.25 \times 10 + 0.75 \times 8$	8.5

Parallel trends holds? **No.**

$\widehat{\delta}_{\text{DiD}} \neq$ ATT because PT bias $= 9.5 - 8.5 = 1.0 \neq 0$

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends
will **mechanically** break parallel trends.

Fix: control for sex

If sex is the only covariate driving differential trends,
then *conditioning on sex* removes the PT bias.

Several ways to do this:

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

We will discuss all four.

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D{=}1,\ X] = \mathbb{E}[\Delta Y(0) \mid D{=}0,\ X]$ for all X

The standard TWFE with additive X

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta\,(D_i \times \text{Post}_t) + \boldsymbol{\beta}\,X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Fatal problem: if X_i is time-invariant, it is collinear with α_i and drops out — whether you use unit fixed effects *or* a saturated dummy specification. An additive, separable X cannot deliver conditional parallel trends if the covariate is absorbed before it can do any work.

What if returns to X change over time?

Example: discrimination against women is not constant — $\beta_{\text{post}} \neq \beta_{\text{pre}}$.

Allow time-varying returns in the DGP:

$$Y_{it}(0) = \alpha_i + \gamma_t + \beta_t X_i$$

Write out the **post** and **pre** equations for unit i:

$$Y_{i,\text{post}}(0) = \alpha_i + \gamma_{\text{post}} + \beta_{\text{post}} X_i$$

$$Y_{i,\text{pre}}(0) = \alpha_i + \gamma_{\text{pre}} + \beta_{\text{pre}} X_i$$

Subtract (post − pre):

$$\Delta Y_i(0) = \underbrace{(\gamma_{\text{post}} - \gamma_{\text{pre}})}_{\Delta\gamma} + \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{\Delta\beta} \cdot X_i$$

Now take expectations by group

Starting from $\Delta Y_i(0) = \Delta\gamma + \Delta\beta \cdot X_i$, condition on treatment status:

$$E[\Delta Y(0) \mid D{=}1] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}1]$$
$$E[\Delta Y(0) \mid D{=}0] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}0]$$

Parallel trends bias:

$$\underbrace{E[\Delta Y(0) \mid D{=}1] - E[\Delta Y(0) \mid D{=}0]}_{\text{PT bias}} \;=\; \Delta\beta \;\cdot\; \big(E[X \mid D{=}1] - E[X \mid D{=}0]\big)$$

PT fails whenever **both**: returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X.

Fix for Bias #1: interact $\mathrm{Post} \times X$

$$Y_{it} = \alpha_i + \lambda_t + \delta\,(D_i \times \mathrm{Post}_t) + \beta\,X_i + \boldsymbol{\gamma}\,(\mathrm{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post (Spec B in the lab).

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

LESSON

Bias #2: Heterogeneous Treatment Effects

What if the effect of college differs by sex?

Previously we worried about **trends in** $Y(0)$ being unequal across groups.

Now the concern is different:

$$\mathbb{E}[Y(1) - Y(0) \mid \text{Male}] \neq \mathbb{E}[Y(1) - Y(0) \mid \text{Female}]$$

Do the rich and the poor get the same return to college?
Do workers in cities and towns get the same return?

If not, **and** the treatment group is imbalanced on these covariates,
the 2×2 DiD will not recover the ATT.

Proof, step 1 — the TWFE model and conditional PT

The TWFE model with one covariate X:

$$Y_{it} \;=\; \alpha_i + \gamma_t + \delta\,(T_t \times D_i) + \beta\,X_i + \varepsilon_{it}$$

Conditional parallel trends says, for each value of X:

$$\mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D{=}1,\ X] \;=\; \mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D{=}0,\ X]$$

Rearranged — the missing counterfactual for the treated is identified:

$$\mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] \;=\; \mathbb{E}[Y_{\text{pre}} \mid D{=}1, X] + \mathbb{E}[Y_{\text{post}} \mid D{=}0, X] - \mathbb{E}[Y_{\text{pre}} \mid D{=}0, X]$$

Proof, step 2 — plug TWFE in: the slopes look the same

From the TWFE model, observed conditional means:

$$\mathbb{E}[Y_{\text{pre}} \mid D{=}1, X] = \alpha_1 + \alpha_3 + \beta\, X$$

$$\mathbb{E}[Y_{\text{post}} \mid D{=}0, X] = \alpha_1 + \alpha_2 + \beta\, X$$

$$\mathbb{E}[Y_{\text{pre}} \mid D{=}0, X] = \alpha_1 + \beta\, X$$

Counterfactual for treated (step 1 identity):

$$\mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \beta\, X$$

Treated outcome, post: $\mathbb{E}[Y(1)_{\text{post}} \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta\, X$

Both potential outcomes carry the *same* β. Subtract: ATT $= \delta$. *Suspiciously clean.*

Proof, step 3 — what if $Y(1)$ and $Y(0)$ have different slopes in X?

Let the slopes differ: write β_1 for the $Y(1)$ slope, β_2 for $Y(0)$:

$$\begin{aligned}\mathbb{E}[Y(1)_{\text{post}} \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta_1\, X \\ \mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 \qquad + \beta_2\, X\end{aligned}$$

True ATT given X:

$$\text{ATT}(X) \;=\; \delta + (\beta_1 - \beta_2)\, X$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on* X
- TWFE fits one β, not two — so $\hat{\delta}$ recovers ATT only if $\beta_1 = \beta_2$

Proof, step 4 — TWFE imposes homogeneous δ in X

Assumption: TWFE-with-X requires the treatment effect to be the same at every value of X.

- If X is **sex**: effect must be equal for men and women
- If X is **income**: effect must be equal at $1 and at $1,000,000
- If X is **age**: effect must be equal at 25 and at 65

OLS doesn't warn you that you're making this assumption. The math does.
If effects actually vary with X, TWFE averages across the variation
and returns a weighted parameter you didn't ask for.

Fix for Bias #2: saturate treatment with X

1. **Run a saturated regression on first differences** — interact treatment with every X
2. **Predict $\Delta Y(1)$ and $\Delta Y(0)$ for each unit** — average the gap in the treated group

```
* 1. Saturated regression on first differences
reg dy i.ever_treated ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.educsq ///
    i.ever_treated#i.marr i.ever_treated#i.nodegree ///
    i.ever_treated#i.black i.ever_treated#i.hisp ///
    i.ever_treated#c.re74 i.ever_treated#i.u74 ///
    $X, robust

* 2. Predict potential outcomes; average the gap in the treated group
replace ever_treated = 1
quietly predict dy_hat_1, xb
replace ever_treated = 0
quietly predict dy_hat_0, xb
replace ever_treated = evt_orig // restore
gen tau_sat = dy_hat_1 - dy_hat_0
quietly sum tau_sat if ever_treated == 1 // ATT = r(mean)
```

Bias #3: time-invariant covariates get absorbed

If you need X_i for conditional PT, but X_i doesn't change over time,
TWFE will **delete it**.

TWFE "demeans" the data. The mean of a constant is itself.
So $X_i - \bar{X}_i \equiv 0$ — it drops out of the regression entirely.

Fix: include $X_i \times \mathrm{Post}_t$ instead. The Post interaction is time-varying,
so it survives demeaning and can do the work of conditioning on X.

Three biases — one family of fixes

Bias	What TWFE imposes	Fix
#1 Trend imbalance	β constant pre and post	$\text{Post} \times X$
#2 Heterogeneous TE	δ same at every X	$D \times X$ (saturate treatment)
#3 Time-invariant X	X_i survives demeaning	$\text{Post} \times X$ (same fix as #1)

Most researchers include additive X and call it a day. That is the wrong spec. To fix all three you need both $\text{Post} \times X$ and $D \times X$. It is rare to see this in published work.

Now: the LaLonde application

Switch to **02b-lalonde.pdf**

We will walk through the code together,
then return here for OR, IPW, and DR.

LESSON

Alternative Estimators

Three more estimators

Now we will look at three more estimators.

These are alternatives to the regression-based specs we just covered.

1. **IPW** — reweight controls to look like treated
2. **Outcome regression (OR)** — model the control trend, impute the counterfactual
3. **Doubly robust (DR)** — combine both; consistent if *either* is right

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

$\hat{p}(X_i) = \widehat{\Pr}(D_i = 1 \mid X_i)$ (propensity score) $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i.
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i))/(1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{\text{ATT}} = \sum w_i \, \Delta Y_i \; / \; n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \to 1$ for a control unit, its weight $\to \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR — or “regression adjustment”

$$\widehat{\mathrm{ATT}}_{\mathrm{OR}} = \frac{1}{n_T} \sum_{D_i=1} \Big[\Delta Y_i - \hat{\mu}_0(X_i) \Big]$$

$\hat{\mu}_0(X_i) = \widehat{E}[\Delta Y \mid D{=}0,\ X]$ estimated on controls only

The estimator is increasingly called **regression adjustment** (RA) in the modern applied literature. Same idea, newer name.

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** ATT $= \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

Surprise — you already know OR

OR step 2: *"Regress ΔY on X using controls only."*

That is exactly what the saturated TWFE does.

Let's prove it slowly.

Proof, step 1 — the saturated model

Spec C runs this regression on first differences:

$$\Delta Y_i = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\delta\, D_i}_{\text{treatment}} + \underbrace{\beta'\, X_i}_{\text{covariates}} + \underbrace{\tau'\,(D_i \times X_i)}_{\text{treatment interactions}} + \varepsilon_i$$

$D_i \in \{0, 1\}$ and **every** element of X_i is interacted with D_i.

Proof, step 2 — separate by group

Plug $D_i = 0$ and $D_i = 1$ into the model:

$$D_i = 0: \quad \Delta Y_i = \alpha + \beta' X_i + \varepsilon_i$$

$$D_i = 1: \quad \Delta Y_i = (\alpha + \delta) + (\beta + \tau)' X_i + \varepsilon_i$$

The two groups have **entirely different** parameters.

Controls contribute to (α, β). Treated contribute to $(\alpha + \delta, \beta + \tau)$.

Proof, step 3 — OLS separates completely

Re-label: $a_0 = \alpha,\ b_0 = \beta,\ a_1 = \alpha + \delta,\ b_1 = \beta + \tau.$

The OLS objective is:

$$\min_{a_0, b_0, a_1, b_1} \underbrace{\sum_{D_i=0} \left(\Delta Y_i - a_0 - b_0' X_i\right)^2}_{\text{only controls}} + \underbrace{\sum_{D_i=1} \left(\Delta Y_i - a_1 - b_1' X_i\right)^2}_{\text{only treated}}$$

The two sums share **no parameters**.

$(\hat{a}_0, \hat{b}_0)$ = OLS of ΔY on X **using controls only.**

Proof, step 4 — Spec C and OR make the same imputation

For a treated unit i, what does each estimator predict as $\Delta Y_i(0)$?

Spec C (saturated TWFE)	set $D_i{=}0$:	$\hat{a}_0 + \hat{b}_0' \, X_i$
OR / regression adjustment	control OLS:	$\hat{a}_0 + \hat{b}_0' \, X_i$

Identical prediction $\Rightarrow$ identical ATT. **Spec C is OR.**

One estimator, two names — just like the 2$\times$2

Recall: six formulas for the 2$\times$2, all giving the same number.

Same principle here.

Saturated TWFE (Spec C) and outcome regression (OR / RA) are **the same estimator** expressed two different ways.

We will confirm this in the do file: both print $1,770.

DR: two chances to be right

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i)\right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\times$	$\checkmark$
$\times$	$\checkmark$	$\checkmark$
$\times$	$\times$	**No**

"Doubly robust" $\neq$ doubly unbeatable.

Lab: ten estimators, one benchmark

`Labs/Lalonde-Covariates/lalonde_all_specs.do`

NSW treated $+$ CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B (Post$\times X$) · Spec C (Saturated)

OR by hand · OR `drdid` · IPW by hand · IPW `drdid` · DR by hand · DR `drdid`

A Monte Carlo to rank the estimators

Estimators:	TWFE, OR, IPW, DR
DGPs:	4 (vary whether OR model and pscore are correct)
n:	1,000 Replications: 10,000
Report:	Bias, RMSE, SE, coverage, CI length

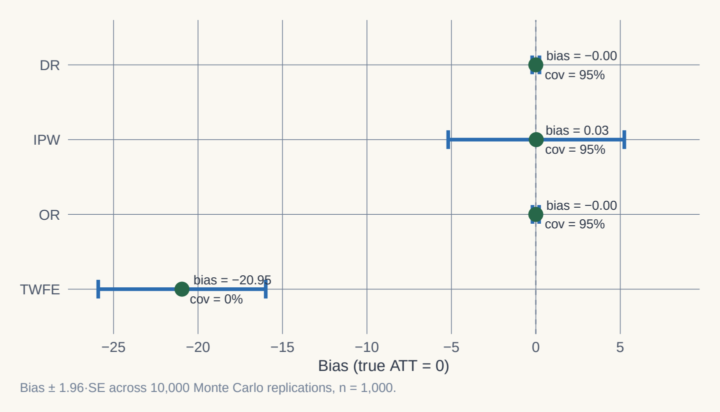
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	-20.95	21.12	2.53	0.000	9.91
OR	-0.001	0.10	0.10	0.950	0.40
IPW	$+0.026$	2.77	2.66	0.952	10.44
DR	-0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

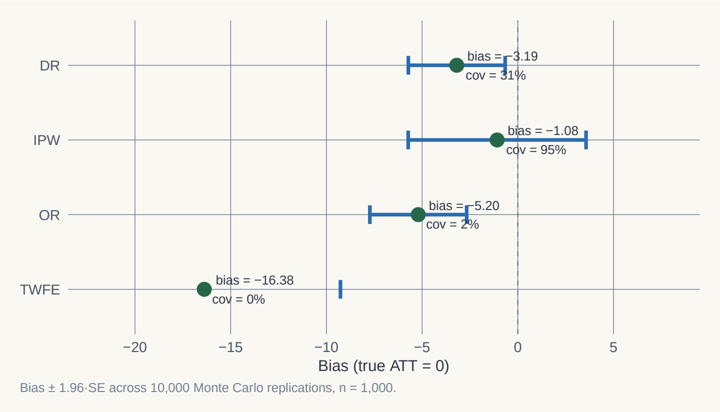


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	-16.38	16.54	3.63	0.000	14.22
OR	-5.20	5.36	1.29	0.015	5.05
IPW	-1.08	2.66	2.37	0.949	9.31
DR	-3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X:** coverage zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when right. Coverage 1.5% when wrong.
- **IPW:** less efficient, but robust to outcome misspecification.
- **DR:** best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—

from **"does PT hold?"**

to **"is my model of $Y(0)$ or $p(X)$ correct?"**

You still have to answer it.

Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect.**

The benchmark: **$1,794**.

LaLonde (1986): the RCT found ≈\$800

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect ≈ **\$800**

A clean causal estimate. The benchmark exists.

LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from $-$**\$16,000 to** $+$**\$3,000**

The true effect was \$800. Not one method recovered it.
"The available evidence calls into serious question the ability of nonexperimental evaluators. . . "

Empirical labor economics faces an evaluation problem.

Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to $1,794

Selective use of the data + right method = nearly recovered the truth.

Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **$1,794**

Let's see which specs recover it — and which don't.

Naive TWFE: $3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

Estimator	ATT	vs benchmark
Naive TWFE	**$3,621**	$+$$1,827

No covariates — the CPS controls trend very differently from NSW treated.

Additive X: still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

Estimator	ATT	vs benchmark
Additive X	**\$3,621**	$+$\$1,827

Adding X as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

Post $\times X$: \$1,711 — now in the right range

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree ///
    i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

Estimator	ATT	vs benchmark
Post $\times X$	**\$1,711**	$-$\$83

Saturated TWFE: $1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

Estimator	ATT	vs benchmark
Saturated TWFE	**$1,770**	−$24

OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

Estimator	ATT	vs benchmark
OR by hand / `drdid reg`	**$1,770**	−$24

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

IPW (Abadie 2005): $1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

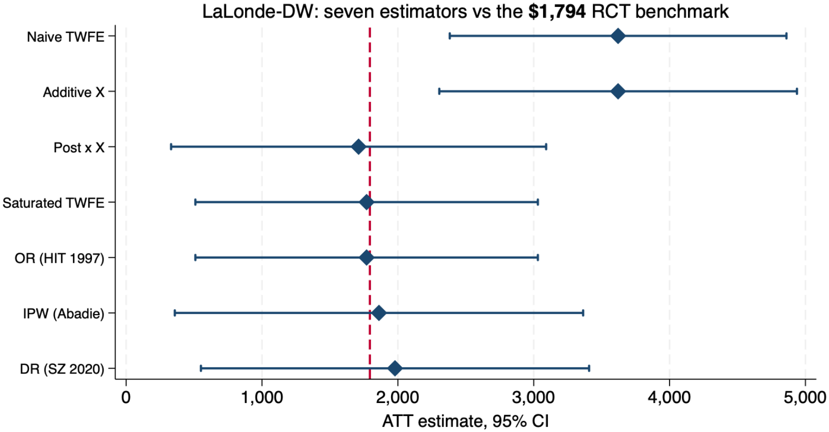
Estimator	ATT	vs benchmark
IPW	**$1,861**	+$67

DR (Sant'Anna–Zhao 2020): $1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Estimator	ATT	vs benchmark
DR	**$1,979**	+$185

All seven estimators vs the $1,794 benchmark



Red dashed line = RCT benchmark ($1,794). Saturated TWFE uses OR's drdid SE.
by-hand and drdid estimates agree to the dollar for OR and IPW.

Covariates are not robustness checks — they are design

- People say: *“if adding covariates changes results, be suspicious.”*
- That gets it backwards.
- We **know** the effect is $1,794.
- Without covariates, or with additive covariates, we get $3,621 — *wrong*.
- The correct specs recover $1,770–$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is never

“did I control for X?”

The question is always

“did I get the model right?”